

Decent Coaching Classes

IX CBSE

Tuesday Holiday Homework

Ch 2: Linear Polynomials | Ch 3: The World of Numbers | Ch 4: Algebraic Identities | Ch 5: Circles

Section A — Chapter 2: Introduction to Linear Polynomials

1. Write the degree of each of the following polynomials:

(a) $6x - 1$ (b) $5x^2 + 3x - 2$ (c) -8 (d) $2y^3 + y - 3$

2. If $p(x) = 5x - 2$, find $p(0)$, $p(3)$ and $p(-2)$.

3. Find the zero of the linear polynomial $p(x) = 7x - 21$.

4. A taxi charges ₹40 as a fixed fare plus ₹12 per km travelled. Write a linear expression for the total fare F for d km, and find the fare for 8 km.

5. Draw the graphs of $y = 3x$ and $y = 3x - 2$ on the same axes and describe the relationship between the two lines.

Section B — Chapter 3: The World of Numbers

6. Classify the following numbers as natural, whole, integer, rational or irrational (a number may belong to more than one group):

(a) -7 (b) 0 (c) $3/5$ (d) $\sqrt{2}$ (e) 4.25

7. Without actual long division, state whether $7/20$ and $4/15$ will have terminating or repeating decimal expansions. Give a reason.

8. Represent $\sqrt{2}$ on the number line using a suitable geometric construction. (Briefly describe the steps.)

9. Is every integer a rational number? Is every rational number an integer? Justify with an example each.

10. Rationalise the denominator of $1/\sqrt{3}$.

Section C — Chapter 4: Exploring Algebraic Identities

11. Using the identity $(a + b)^2 = a^2 + 2ab + b^2$, expand $(4x + 7y)^2$.

12. Using the identity $(a - b)^2 = a^2 - 2ab + b^2$, expand $(9p - 2q)^2$.

13. Using the identity $a^2 - b^2 = (a + b)(a - b)$, evaluate 102×98 without direct multiplication.

14. Expand $(x + 6)(x + 5)$ using the identity $(x + a)(x + b) = x^2 + (a + b)x + ab$.

15. Factorise: $16x^2 - 49y^2$

Section D — Chapter 5: I'm Up and Down, and Round and Round (Circles)

16. Define the following terms with a rough figure: (a) radius (b) chord (c) minor arc (d) major arc.

17. State the theorem relating equal chords of a circle and their distances from the centre.

18. A chord of length 24 cm is drawn in a circle of radius 13 cm. Find the distance of the chord from the centre.

19. An arc of a circle subtends an angle of 80° at the centre. What angle does it subtend at a point on the remaining part of the circle?

20. PQRS is a cyclic quadrilateral in which $\angle P = 95^\circ$. Find $\angle R$, and state the property used.

ANSWER KEY

Section A — Linear Polynomials

1. (a) 1 (b) 2 (c) 0 (d) 3
2. $p(0) = -2$, $p(3) = 13$, $p(-2) = -12$
3. $7x - 21 = 0 \Rightarrow x = 3$
4. $F = 40 + 12d$; for $d = 8$, $F = 40 + 96 = ₹136$
5. Both lines have the same slope 3 (parallel lines); $y = 3x$ passes through the origin, while $y = 3x - 2$ is shifted 2 units downward (y -intercept $= -2$).

Section B — The World of Numbers

6. (a) -7 : Integer, Rational (b) 0: Whole, Integer, Rational (c) $\frac{3}{5}$: Rational (d) $\sqrt{2}$: Irrational (e) 4.25: Rational
7. $\frac{7}{20} = \frac{7}{(2^2 \times 5)}$: denominator has only factors 2 and 5 $\Rightarrow$ terminating decimal (0.35). $\frac{4}{15} = \frac{4}{(3 \times 5)}$: denominator has a factor 3 (other than 2, 5) $\Rightarrow$ repeating decimal (0.2666...).
8. Draw a unit square OABC with $OA = AB = 1$ unit on the number line. By Pythagoras theorem, $OB = \sqrt{(1^2 + 1^2)} = \sqrt{2}$. With O as centre and OB as radius, draw an arc cutting the number line — the point where it cuts represents $\sqrt{2}$.
9. Every integer is a rational number (e.g., $-3 = -3/1$), but not every rational number is an integer (e.g., $\frac{3}{5}$ is not an integer).
10. $\frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{3}}{3}$

Section C — Algebraic Identities

11. $(4x + 7y)^2 = 16x^2 + 56xy + 49y^2$
12. $(9p - 2q)^2 = 81p^2 - 36pq + 4q^2$
13. $102 \times 98 = (100+2)(100-2) = 100^2 - 2^2 = 10000 - 4 = 9996$
14. $(x + 6)(x + 5) = x^2 + 11x + 30$
15. $16x^2 - 49y^2 = (4x)^2 - (7y)^2 = (4x + 7y)(4x - 7y)$

Section D — Circles

16. Radius: the distance from the centre to any point on the circle. Chord: a line segment joining two points on the circle. Minor arc: the smaller of the two arcs formed by two points on a circle. Major arc: the larger of the two arcs.
17. Equal chords of a circle (or of congruent circles) are equidistant from the centre, and chords equidistant from the centre are equal.
18. Half-chord $= 12$ cm. Distance $= \sqrt{(\text{radius}^2 - \text{half-chord}^2)} = \sqrt{(13^2 - 12^2)} = \sqrt{(169 - 144)} = \sqrt{25} = 5$ cm.
19. The angle at the centre is double the angle at the circumference on the same arc, so the angle at the remaining part of the circle $= 80^\circ \div 2 = 40^\circ$.
20. Opposite angles of a cyclic quadrilateral are supplementary, so $\angle R = 180^\circ - 95^\circ = 85^\circ$.